

experience—to ensure these factors do not drive our estimates (e.g., adopters being younger and naturally on upward earnings trajectories). Second, we leverage our panel data to implement a difference-in-differences approach indexed to November 2022, the release date of ChatGPT. This allows us to control for time-invariant differences between workers (e.g., high-ability adopters who would have earned more regardless) and examine whether adopters experienced differential changes after the launch of ChatGPT. Crucially, the event-study design allows us to assess whether adopters were on distinct labor market trends before AI chatbots became available. Finally, we verify that our conclusions about the effects of worker-level use are robust when we rely only on the variation in chatbot use that stems from employer policies. Employer initiatives may be more plausibly unrelated to individual worker trajectories, as they reflect firm-level choices rather than worker self-selection into chatbot use.

A third challenge is timing: When should we date the arrival of AI chatbots, and when should we expect their effects on labor market outcomes to appear? We use the launch of ChatGPT in November 2022 as a before-and-after moment in an event-study design, marking a sharp rise in public awareness of AI chatbots.¹¹ We also separately examine effects for early adopters, identified as workers already using chatbots in our 2023 survey round, to allow more time for effects to materialize.¹² Even with rapid adoption, however, labor market effects may take time to materialize. We examine these adjustment dynamics in two ways. First, we implement a dynamic difference-in-differences design to trace how effects unfold monthly following the introduction of AI chatbots. Second, we analyze heterogeneity across occupations that vary in the flexibility of labor rules.

A final concern is that difference-in-differences identifies only *differential* effects for

¹¹This timing is motivated by the rapid uptake of ChatGPT among workers: most adopters began using it within a year of its launch (Humlum and Vestergaard, 2025). Appendix E.2 shows that ChatGPT remains the dominant AI chatbot to date. Moreover, event studies centered on its launch in November 2022 have become common in the literature (e.g., Brynjolfsson, Chandar and Chen (2025); Eisfeldt et al. (2024); Lichtinger and Hosseini Maasoum (2025); Schubert (2025); Teutloff et al. (2025)), facilitating comparison across studies.

¹²Our survey does not record the exact timing of adoption, precluding event studies indexed to workers' month of first use. However, even if such information were available, adoption is unlikely to represent a sharp before-and-after event, as workers typically ramp up usage gradually. This is another reason why we expect difference-in-differences effects to emerge over time rather than immediately.

adopters, leaving any level effect on non-adopters unidentified—the “missing intercept.” For example, when we compare earnings relative to non-encouraged non-adopters in Section 3.2, any general equilibrium effects that also affect these workers are differenced out. Indeed, Section 2.3 showed that AI chatbots have created new job tasks even for non-adopters. The key question for our analysis below is whether such spillovers also surface in earnings or recorded hours. For instance, adopting firms might expand by poaching workers from non-adopting competitors, biasing our workplace employment estimates upward. Alternatively, if firms adjust labor demand in anticipation of AI chatbots—rather than based on their actual adoption—the estimates would be biased toward zero. We assess the potential magnitude of these spillover effects from several complementary angles.

First, we assess potential spillovers at different levels of aggregation by estimating difference-in-differences models at the worker, workplace, and occupation levels. For instance, the fact that encouraged workplaces do not experience differential changes in labor demand in Section 3.4 suggests that spillovers in employment outcomes within workplaces are limited. Consistent with this interpretation, Figure F.3 shows that non-adopters in encouraged workplaces have fared similarly to their counterparts in non-encouraged workplaces following the introduction of AI chatbots. Extending this approach, Appendix D.1 shifts the analysis to the occupation level, showing that our exposed study occupations have also evolved similarly to the rest of the economy, maintaining a stable 20% share of aggregate hours and earnings before and after the arrival of AI chatbots.

Second, Appendix A.2.6 uses a market equilibrium model to show that our difference-in-differences estimates are informative about the magnitude of the “missing intercept.” In particular, under reasonable labor demand elasticities, null effects at both the workplace and worker levels imply that the occupation-wide equilibrium effect is also negligible. The key intuition is that adopters generate spillovers to non-adopters only when their own market outcomes change. Thus, the absence of adoption effects at both levels implies that spillovers to non-adopters are likewise limited.

Finally, as discussed in Section 3.5, workers’ perceived earnings impacts offer an independent lens for evaluating general equilibrium spillovers. In particular, the fact that virtually *all* non-adopters report no effect on their earnings further reinforces the finding that spillover impacts of AI chatbots remain minimal.

Taken together, these different analyses—difference-in-differences at varying levels of aggregation, equilibrium calibration of micro-level estimates, and perceived impacts among workers—all point to the conclusion that AI chatbots have not had any meaningful impact on non-adopters’ earnings or employment thus far.

3.1.2 Regression Specifications

Sampling Design and Data. Our data and sampling are designed to answer the questions described above. By linking survey responses from workers in the most highly exposed occupations to comprehensive matched employer–employee panel data covering the entire Danish economy, we can study labor market outcomes before and after the arrival of AI chatbots. Crucially, the matched data allow us to track not only the surveyed workers themselves but also other workers at the same workplaces, including those who have since left.

Our survey sampling design also imposes restrictions that we explicitly account for in the analysis. In particular, because we survey workers in exposed occupations, all respondents are, by construction, employed in those occupations at the time of the survey sampling. Accordingly, all specifications include survey-occupation $\times$ time fixed effects, ensuring that all comparisons are made within survey strata and time period, between workers (or workplaces) that differ in their adoption of AI chatbots. We also study heterogeneity in effects by survey occupation.

Let Y_{it} denote a labor market outcome (e.g., earnings) for worker i in month-year t , let X_i denote survey-occupation fixed effects and worker pre-determined characteristics—age, gender, and potential experience in 2022—and let A_i indicate whether the worker has adopted AI chatbots. We examine both worker-level effects (e.g., worker earnings by

individual usage) and workplace-level effects (e.g., workplace employment by employer initiatives).

Dynamic Difference-in-Differences. To assess how impacts evolve over time, we employ a dynamic difference-in-differences specification that traces the monthly evolution of outcomes for adopters relative to non-adopters around the introduction of AI chatbots:

$$Y_{it} = \underbrace{\sum_{\tau} \lambda_{1\tau} X_i \mathbf{1}_{\{t=\tau\}}}_{\text{Controls}} + \underbrace{\sum_{\tau \neq 2022M11} \beta_{\tau} A_i \mathbf{1}_{\{t=\tau\}}}_{\text{Dynamic Diff-in-Diffs}} + \alpha_i + \varepsilon_{it}, \quad (2)$$

where α_i are individual (workplace) fixed effects. The parameters of interest, β_{τ} , capture the differential changes in labor market outcomes for adopters, indexed to November 2022, the release date of ChatGPT. We estimate Equation (2) by OLS and cluster the standard errors at the survey workplace-occupation level.

Pooled Difference-in-Differences. To examine heterogeneous impacts across multiple dimensions, we also estimate a pooled specification comparing average outcomes of adopters and non-adopters before and after the introduction of AI chatbots. To capture dynamics, we estimate separate effects for each year following the launch of ChatGPT:¹³

$$Y_{it} = \underbrace{\sum_{\tau} \lambda_{1\tau} X_i \mathbf{1}_{\{t=\tau\}}}_{\text{Controls}} + \underbrace{\lambda_2 A_i t}_{\text{Time Trends}} + \underbrace{\sum_{y=2023}^{2024} \beta_y A_i \mathbf{1}_{\{y(t)=y\}}}_{\text{Pooled Diff-in-Diffs}} + \alpha_i + \varepsilon_{it}, \quad (3)$$

where adopter-specific time trends guard against spurious pre-trends driving the estimates (which the dynamic specification instead reveals nonparametrically). The parameters of interest, β_y , measure how labor market outcomes for adopters changed in year y after the introduction of AI chatbots. We estimate Equation (3) by OLS and cluster the standard errors at the survey workplace-occupation level.

¹³To align with our dynamic difference-in-differences specification (Equation (2)), which defines post-periods as months after 2022M11, we include 2022M12 in the 2023 estimate.